Case study report



UNIVERSITY OF SKÖVDE

Autonomous Mobile Robot Simulation for Industrial Material Handling

Project Report

Course: Autonomous Systems VP718A

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Student: Maritza Elizabeth Albán Escobar

Course responsible: Dr. Richard Senington

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1 Introduction

Autonomous mobile robots (AMRs) are increasingly being considered for industrial intralogistics, where the movement of materials between storage, production, and dispatch areas is an important part of warehouse operations. Their potential benefits include greater flexibility in material transport, improved productivity, and the ability to adapt to changing operational conditions. However, these benefits depend on the integration of reliable localization, environmental perception, motion control, task planning, and safety mechanisms (Fragapane et al., 2022; Lackner et al., 2024).

There are an important distinction between automation and autonomy. An automated vehicle may execute predefined routes and schedules with limited human involvement, but this does not necessarily mean that it can respond independently to unexpected changes in its environment. Autonomous systems use information from their surroundings to estimate their state and support decisions about navigation and task execution. Franke et al. (2026) also highlights the importance of evaluating the capabilities a system demonstrates in practice rather than classifying it as autonomous based on its intended function alone.

This distinction is relevant to the warehouse scenarios considered in this project. The first company relies on human-operated vehicles for material transport, while the second uses robots that follow predefined routes at scheduled times and must be switched off when a person enters the environment. Human-operated transport offers flexibility through human judgement, whereas route-based automation can provide repeatable operation but may be less adaptable. A mobile robotic solution that combines localization, navigation, task execution, and responses to dynamic obstacles could potentially address some of the limitations of these approaches. However, the suitability of such a solution depends on its demonstrated performance and its ability to operate safely in the intended environment (Lackner et al., 2024).

The aim of this project is to investigate the capabilities and limitations of a simulated AMR system developed in PyBullet. The implementation combines motion control, simulated LiDAR sensing, particle-filter-based localization, waypoint navigation, multiple robots executing different jobs, and a safety response to potential dynamic obstacles. Localization is a critical component of mobile robot navigation because the robot needs an estimate of its position and orientation to move through its environment and complete assigned tasks (Huang et al., 2023). However, the reliability of a localization estimate is also important for safe operation, particularly when navigation decisions may affect nearby people or other moving objects (Hafez et al., 2024). The simulation is therefore used to examine task execution and the robots' responses when potential obstacles are detected, while recognizing the limitations of the implemented mechanisms.

Finally, the report evaluates the autonomy level demonstrated by the implementation using selected autonomy assessment models and considers its potential application in industrial warehouses. It also discusses the limitations of the simulation and the safety concerns that would need to be addressed before deployment in an environment shared with human workers. The evaluation considers not only whether robots can execute their assigned tasks, but also how reliably they do so and whether their responses to dynamic obstacles provide sufficient evidence of safe operation.

2 Problem Specification and Scope

2.1 Problem Definition

Two companies seek to improve warehouse material handling. The first relies on human-operated vehicles, while the second uses automated vehicles that must stop whenever people enter the operating area. Both need more flexible automation that allows vehicles and people to share the warehouse environment. This project explores the use of a common AMR solution through simulation.

2.2 Autonomy Assessment Framework

Two models are considered for evaluating autonomy and automation: the framework proposed by Franke et al. (2026) and the model developed by Parasuraman et al. (2000). Although both address different aspects of automation, they provide complementary approaches to assessing system capabilities.

Franke et al. (2026) propose a six-level autonomy maturity framework, ranging from Zero Autonomy to Independence. It evaluates mobile robot capabilities through four dimensions: decision-making, awareness, flexibility and system behavior.

Parasuraman et al. (2000) propose a model that examines automation across four functional stages: information acquisition, information analysis, decision and action selection, and action implementation. Each function can operate at a different level of automation, depending on the degree of human or automated involvement.

Table 1 summarizes the main characteristics of both models and their relevance to this project.

Table 1. *Comparison of autonomy assessment models*

Criterion	Franke et al. (2026)	Parasuraman et al. (2000)
Main focus	Autonomy maturity of mobile robots	Levels of human interaction with automation
Assessment approach	Six maturity levels	Different automation levels for four functions
Main dimensions	Decision-making, awareness, flexibility and system behavior	Information acquisition, information analysis, decision/action selection and action implementation
Application	Mobile robots in warehousing	General automated systems
Main contribution	Evaluates the progression towards independent robot operation	Evaluates how responsibilities are distributed between humans and automation
Relevance to this project	Directly applicable to warehouse AMRs	Useful for understanding human intervention in automated functions

Franke et al. (2026) is selected as the primary assessment framework because it specifically addresses the autonomy maturity of mobile robots in warehouse environments. Its six maturity levels and descriptive dimensions provide a direct basis for evaluating the capabilities demonstrated by the companies and the proposed

AMR system. Although Parasuraman et al. (2000) offers a useful approach to understanding human involvement in automated functions, its focus is broader and does not directly classify the autonomy maturity of mobile robots. Therefore, Franke et al. (2026) is considered more suitable for the objectives and application context of this project.

2.3 Assessment of the Companies' Current Autonomy Levels

The autonomy maturity framework presented in Section 2.2 is applied to assess the current material-handling systems of both companies. The assessment considers the four dimensions proposed by Franke et al. (2026): decision-making, awareness, flexibility and system behavior. Table 2 summarizes the observed characteristics of each system and their correspondence with the framework.

Table 2. *Comparative Autonomy Assessment of the Two Companies*

Assessment dimension	Company 1	Company 2
Decision-making	Human driver makes operational decisions.	Executes predefined routes and schedules.
Awareness	Environmental awareness provided by the driver.	Stops when a person enters the operating area; detection mechanism unspecified.
Flexibility	Human driver can adapt to changing conditions.	Limited by predefined routes and stopping conditions.
System behavior	Entirely human-operated vehicle.	Automated transport with predefined instructions.
Assigned autonomy level	Level 0 – Zero Autonomy	Level 1 – Safe and Simple Purpose Fulfillment (provisional)

As shown in Table 2, Company 1 relies on human judgement for navigation, environmental awareness and operational decisions. Although the driver can respond flexibly to changing conditions, these capabilities are not performed autonomously by the vehicle. Therefore, the system is classified at Level 0 – Zero Autonomy.

Company 2 demonstrates a higher degree of automation because its vehicles execute predefined transport operations without continuous manual driving. However, the available information does not demonstrate independent localization, mapping or adaptive navigation. The requirement to stop when a person enters the operating area also indicates limited operational flexibility. Therefore, the system is provisionally classified at Level 1 – Safe and Simple Purpose Fulfillment.

2.4 Project Objectives and Requirements

The requirements and acceptance criteria are summarized in Table 3. They establish the main capabilities needed to evaluate the proposed concept.

Table 3. *System requirements and acceptance criteria.*

ID	Requirement	Objective	Evaluation criterion
R1	Localization	Estimate vehicle position and orientation.	Measure localization accuracy.
R2	Environment perception	Detect obstacles and unexpected objects.	Evaluate obstacle detection in defined scenarios.
R3	Human identification	Identify people as a specific obstacle category.	Identify whether people can be distinguished from other obstacles.
R4	Navigation	Move between warehouse locations.	Verify destination and waypoint completion.
R5	Task planning	Organize material transport efficiently.	Compare task completion and travel cost.
R6	Multi-robot operation	Allow multiple robots to execute different jobs.	Evaluate task execution and coordination.
R7	Operational safety	Respond to potential obstacles.	Evaluate CLEAR, SLOW and STOP responses.
R8	Operational flexibility	Recover from temporary route obstructions.	Evaluate waiting, replanning and navigation recovery.
R9	Performance evaluation	Assess overall system performance.	Record quantitative experimental results.

2.5 Scope and Assumptions

The project focuses on simulating an autonomous material handling system in a warehouse environment, including localization, navigation, task execution, multiple robots and interactions with people and obstacles. The study is limited to a simulated environment and does not include the physical development or deployment of AMRs. Real-world sensor performance, industrial safety certification and integration with existing warehouse infrastructure are outside its scope.

3 Proposed Solutions and Design Selection

Two design alternatives were considered to address the warehouse automation requirements. Table 4 summarizes their main features and differences. Proposal A follows a conventional navigation approach and reflects the system implemented in the PyBullet simulation, while Proposal B explores more advanced techniques that could provide greater autonomy and adaptability but require additional development.

Table 4. *Design alternatives and preliminary selection.*

Function	Proposal A: Conventional AMR	Proposal B: Advanced AMR
Mobile platform	Differential drive	Differential or omnidirectional drive
Localization (R1)	Particle filter using odometry and a known map	SLAM for simultaneous localization and mapping
Perception (R2)	LiDAR-based obstacle detection	Camera-based or camera-LiDAR perception
Human identification (R3)	Geometric obstacle detection without person classification	Visual person recognition using computer vision

Navigation (R4)	A* path planning and waypoint navigation	Reinforcement Learning-based navigation
Task planning (R5)	FIFO task assignment and route-based planning	Cost-based task allocation and optimization
Multi-robot operation (R6)	Multiple robots executing assigned tasks	Coordinated multi-robot planning and task allocation
Safety (R7)	Speed reduction, stopping and path replanning	Context-aware obstacle avoidance and adaptive navigation
Recovery (R8)	Resume navigation or replan after an obstruction	Adaptive recovery and decision-making
Evaluation (R9)	Quantitative simulation tests	Simulation and experimental validation of advanced methods

Proposal A was selected because it provides a feasible implementation within the available simulation environment. It combines localization using a known map, LiDAR-based obstacle detection, A* path planning, waypoint navigation and multi-robot task execution.

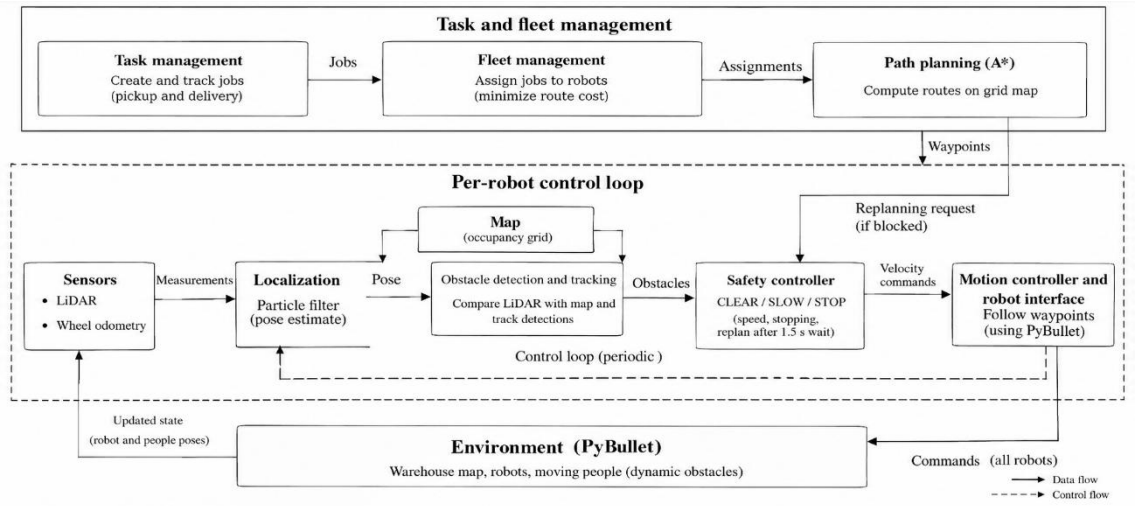
Proposal B represents an alternative based on more advanced perception, localization and decision-making techniques. Although it could offer greater adaptability to chang-ing environments, its implementation would require additional computational re-sources, training and validation.

4 Implementation

4.1 High-Level Architecture

The system is implemented in PyBullet as a multi-robot warehouse simulation. Its main components include task management, fleet coordination, path planning, localization, dynamic obstacle detection and motion control. Figure 1 illustrates the interaction between these components and the main simulation loop.

Figure 1. *Functional architecture of the multi-robot simulation.*



4.2 Localization and Navigation

The robots use a predefined occupancy grid map for navigation and localization. Differential-drive odometry estimates robot motion, while a particle filter combines odometry and LiDAR measurements to estimate position and orientation. The A* algorithm calculates global routes, which are converted into waypoints for the motion controller. A* is also used to calculate alternative routes when replanning is required. The system relies on a known map and does not implement SLAM.

4.3 Multi-Robot Task Management

The simulation includes two robots and four pickup-and-delivery jobs. The task manager maintains job information, while the fleet manager evaluates possible assignments based on route costs. In the selected optimization mode, assignments aim to minimize the maximum route cost across the robots, balancing their workloads. Each robot then executes the waypoints associated with its assigned jobs.

4.4 Dynamic Obstacle Detection and Safety Control

The obstacle detection mechanism compares actual LiDAR measurements with expected distances from the static map. Significant discrepancies are treated as potential dynamic obstacles and grouped into detections, which are tracked across consecutive observations. This approach does not identify people specifically and may produce false positives.

The safety controller uses three states: CLEAR, SLOW and STOP. Depending on the obstacles detected, the robot maintains normal speed, reduces its speed or stops. When a planned route remains blocked, a waiting period of 1.5 seconds is applied before initiating path replanning using A*. This allows the system to respond to obstacles while avoiding immediate route changes caused by temporary detections.

4.5 Control Logic and Autonomy

The robots operate autonomously at task level: they estimate their positions, execute assigned jobs, follow planned routes and react to potential obstacles without continuous human control. However, their autonomy is limited using a predefined map and assigned tasks, as well as the limitations of the obstacle detection mechanism. The safety response provides speed reduction, stopping and conditional path replanning, but does not guarantee complete obstacle avoidance or reliable human identification. The implementation therefore demonstrates a task-level autonomous mobile robot system in a structured simulated environment, rather than a fully autonomous solution ready for unrestricted operation alongside human workers.

The system demonstrates autonomy because the robots can estimate their positions, plan routes, execute assigned tasks, and respond to potential obstacles without continuous human control. Its autonomy is nevertheless limited using a

predefined map and jobs, as well as a safety mechanism that can slow or stop the robots but cannot perform complete obstacle avoidance. Therefore, the implementation demonstrates a task-level autonomous mobile robot system in a structured, simulated environment, rather than a fully autonomous solution suitable for unrestricted operation alongside human workers. This assessment is based on the functions implemented; the reliability of these capabilities must be established through experimental evaluation.

5 Results and Evaluation

5.1 Localization Performance

Several experiments were conducted to evaluate the Particle Filter (PF) under different motor-noise levels and route configurations. The initial test used 50 particles without noise, while all subsequent tests used 80 particles with 5% or 10% motor noise.

Table 5 presents the position RMSE for both robots, comparing the odometry and Particle Filter estimates across the experiments.

Table 5. *Position RMSE (m)*

Experiment	Particles	Noise	Robot 1	Robot 2
Initial test (Scenario 2)	50	0%	0.161	1.223
Scenario 1	80	5%	0.035	0.065
Scenario 2	80	5%	0.069	0.082
High noise (Scenario 2)	80	10%	0.096	3.747
Movement optimization (Scenario 2)	80	5%	0.050	0.054

Similarly, Table 6 shows the orientation RMSE, allowing the accuracy of the estimated robot heading to be evaluated.

Table 6. *Orientation RMSE (degrees)*

Experiment	Robot 1	Robot 2
Initial test (Scenario 2 and 50 particles)	1.84°	12.71°
Scenario 1	1.24°	1.57°
Scenario 2	1.47°	1.63°
High noise (Scenario 2)	1.23°	11.97°
Movement optimization (Scenario 2)	1.15°	1.61°

As shown in Tables 5 and 6, the PF generally improves localization accuracy compared with odometry, particularly in position estimation under 5% motor noise. However, performance degrades significantly for Robot 2 under 10% noise. These results highlight the importance of sensor-based corrections in reducing odometry drift, while also showing that localization performance depends on operating conditions.

5.2 Path Planning and Task Assignment

The path planning and task assignment system was evaluated in two warehouse scenarios using two optimization strategies: minimizing execution time and minimizing the number of movements.

Figure 2 illustrates the resulting robot routes and task assignments. In Scenario 1 (Figure 2a), the tasks are distributed between the two robots, with each robot assigned two jobs. Scenario 2 (Figure 2b) uses the time-based optimization strategy and maintains the same task distribution. In contrast, when movement-based optimization is applied (Figure 2c), Robot 1 is assigned only Job A, while Robot 2 performs Jobs B, C and D. This demonstrates how the assignment strategy adapts to the selected optimization criterion.

Figure 2. *Robot routes and task assignments under different warehouse scenarios and optimization strategies.*

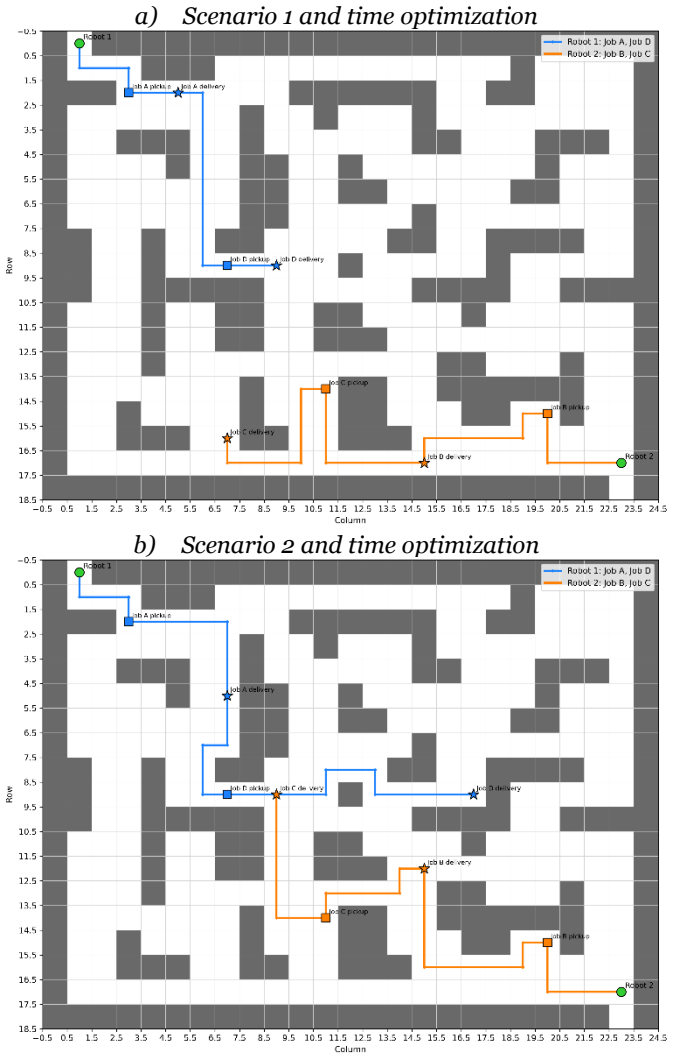


Figure 2. *Robot routes and task assignments under different warehouse scenarios and optimization strategies (Continuation)*

c) Scenario 2 and movement optimization

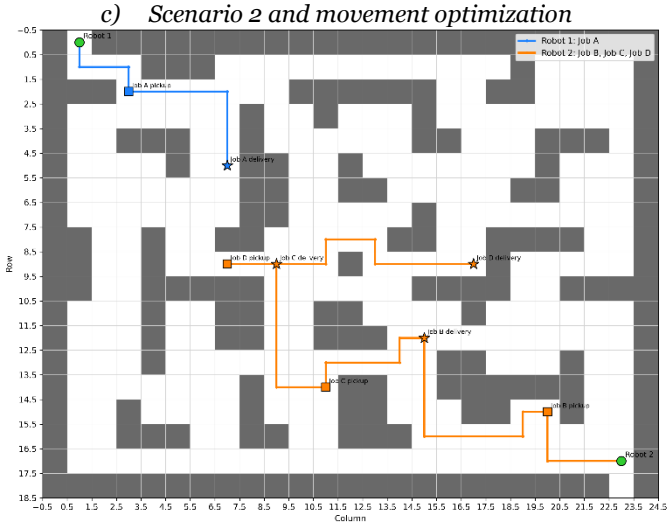


Table 7 compares the execution times, planned movements and task completion across the three experiments.

Table 7. *Path planning and task assignment results*

Scenario	Optimization	Robot 1 / Robot 2 movements	Total movements	Simulation time (s)	Real time (s)	Completion
Scenario 1	Time	17 / 27	44	139.22	300.28	20/20
Scenario 2	Time	29 / 28	57	155.48	419.85	26/26
Scenario 2	Movements	11 / 42	53	221.08	584.23	24/24

Both optimization strategies successfully completed all assigned tasks. In Scenario 2, movement-based optimization reduced the total number of planned movements from 57 to 53 (approximately 7%). However, this came at the cost of a longer execution time. This indicates a trade-off between reducing movements and minimizing execution time.

Overall, the results demonstrate that the system can adapt its task allocation and path planning to different warehouse layouts and optimization objectives.

5.3 Dynamic Obstacle Detection and Response

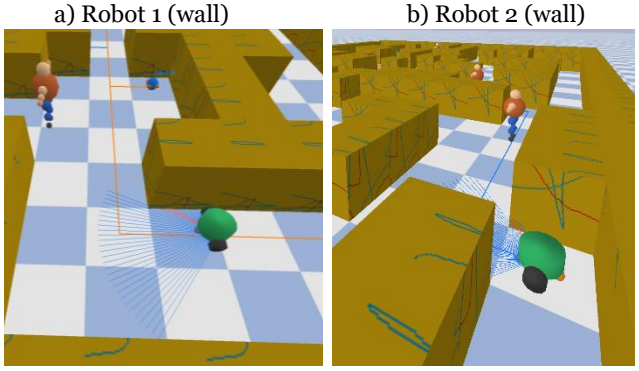
5.3.1 Dynamic obstacle detection

The multi-robot system incorporates LiDAR-based dynamic obstacle detection by comparing real measurements with expected distances from the static map. Potential obstacles are identified and tracked across consecutive observations. As illustrated in Figure 3, the experiments revealed false positives, particularly

near walls and in narrow corridors, where localization errors and measurement discrepancies affected detection reliability.

Figure 3. *False-positive detections during dynamic obstacle monitoring*

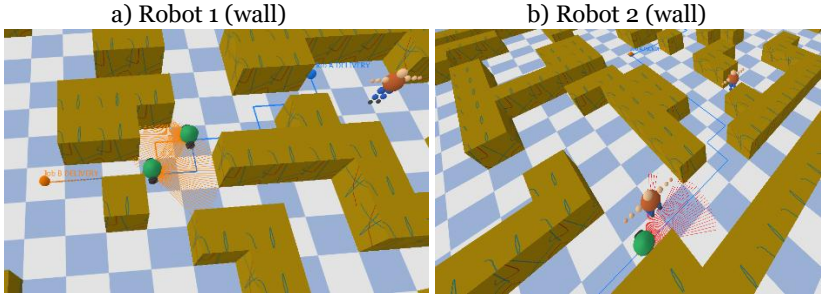
a) Robot 1 (wall) b) Robot 2 (wall)



Despite these limitations, the system was also able to identify moving objects in the robot's field of view. Figure 4 presents examples of true-positive detections, demonstrating that the proposed method can identify objects that are absent from the static map.

Figure 4. *True-positive detections during dynamic obstacle monitoring*

a) Robot 1 (wall) b) Robot 2 (wall)



5.3.2 Safety response and execution results

Once a potential obstacle is detected, the robot adjusts its motion according to the estimated obstacle distance. Three safety states are implemented: CLEAR, SLOW and STOP, allowing the robot to maintain normal speed, reduce its speed or temporarily stop, respectively. Table 8 summarizes the execution results obtained in the two dynamic obstacle experiments.

Table 8. *Dynamic obstacle experiment execution results*

Metric	Scenario 3	Scenario 4
Simulation time (s)	141.08	128.55
Real execution time (s)	344.34	300.39
Robot 1 waypoints completed	10/10	6/6
Robot 2 waypoints completed	2/2	1/1
Total waypoints completed	12/12	7/7
Robot failures	0	0

Both experiments successfully completed all assigned waypoints without recording robot failures. The safety-state transitions recorded during execution are presented in Table 9.

Table 9. *Recorded safety-state transitions during dynamic obstacle experiments*

Safety state	Scenario 3	Scenario 4
CLEAR	40	73
SLOW	17	13
STOP	24	62
Total transitions	81	148

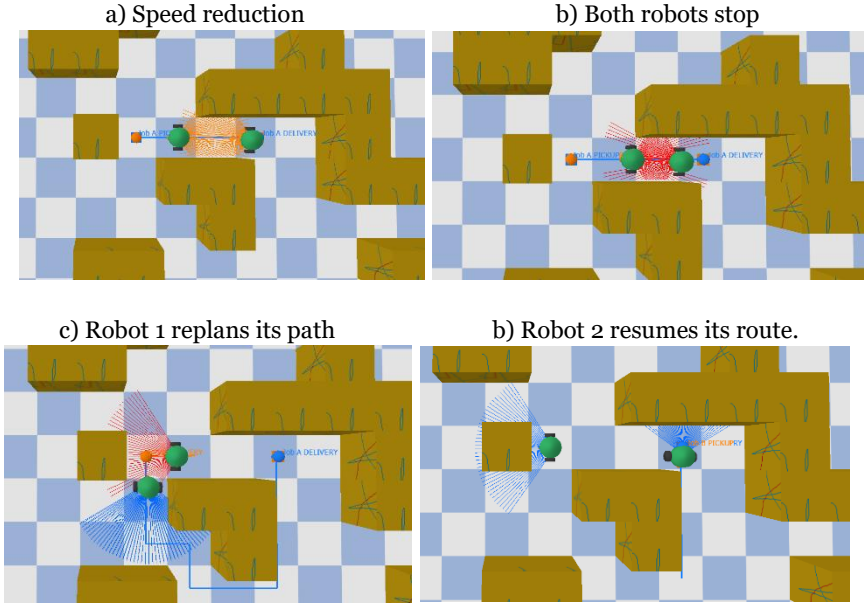
These results confirm that the safety mechanism was activated during navigation, producing both speed reductions and temporary stops in response to potential obstacles.

5.3.3 Robot response and path replanning

In addition to speed adaptation and emergency stopping, an A*-based path replanning mechanism was incorporated to allow robots to calculate alternative routes when their planned trajectories become blocked. A waiting period of 1.5 seconds was introduced before initiating replanning, avoiding immediate route changes in response to temporary obstacles.

Figure 5 illustrates the sequence of robot responses during an obstacle encounter, including speed reduction, stopping, path recalculation and the continuation of navigation.

Figure 5. *Robot response and path replanning in the presence of obstacles*



The experiments demonstrated the feasibility of integrating dynamic obstacle monitoring and response into the multi-robot navigation architecture.

However, detection reliability and false-positive events remain important aspects for further improvement.

6 Practical Application

6.1 Application to the Two Companies

The proposed AMR system offers a potential approach to improving warehouse material handling by increasing operational flexibility and reducing reliance on predefined routes and continuous human control.

For Company 1, the system could reduce dependence on manually operated transport vehicles by introducing autonomous navigation and task execution. For Company 2, it could provide greater flexibility through route planning, obstacle detection and replanning instead of relying exclusively on predefined routes. In both cases, the proposed solution aims to provide task-level autonomy, allowing robots to localize themselves, execute assigned jobs and respond to potential obstacles.

6.2 Environmental Complexity and Target Autonomy

The simulated warehouse provides a structured but challenging environment, including static obstacles, predefined navigation routes, two robots operating simultaneously and four pickup-and-delivery tasks. Dynamic obstacles introduce unexpected changes that require the robots to adjust their movements and potentially replan their routes. This combination allows the evaluation of navigation, task allocation, multi-robot operation and obstacle response, demonstrating the potential usefulness of AMRs in warehouse material handling.

The target autonomy is based on *Level 2 – Orientation and Interaction*, as defined by Franke et al. (2026). This level is selected because the project aims to demonstrate independent localization, navigation and material-handling operations. However, the implementation uses a predefined map rather than performing simultaneous localization and mapping (SLAM). Therefore, the project targets the main operational capabilities associated with this level within a known environment, rather than claiming to fulfil all its descriptors.

6.3 Evaluation of the Achieved Autonomy

The achieved autonomy is evaluated by comparing the target capabilities with the experimental results. Table 10 summarizes the evidence across the four dimensions of the framework proposed by Franke et al. (2026).

Table 10. *Evaluation of the proposed and achieved AMR autonomy*

Dimension	Target capability	Experimental evidence	Assessment
Decision-making	Independent task execution and route planning.	Two robots completed assigned jobs using task allocation and A*.	Demonstrated within predefined tasks.
Awareness	Localization and environmental perception.	Particle Filter localization and LiDAR obstacle detection.	Partially demonstrated; high-noise errors and false positives occurred.
Flexibility	Respond to obstacles and recover through replanning.	CLEAR, SLOW and STOP responses; all waypoints completed in Scenarios 3 and 4.	Demonstrated in tested scenarios, with limitations.
System behavior	Independent multi-robot task execution.	Two robots completed different jobs across the task-planning experiments.	Demonstrated in simulation.

The results demonstrate that the robots can localize themselves, navigate using A* and execute assigned material-handling tasks. However, localization reliability decreases under high-noise conditions, with Robot 2 presenting a position RMSE of 3.747 m and an orientation RMSE of 11.97°. Dynamic obstacle experiments showed that the robots could reduce speed, stop and replan their routes, completing all waypoints in Scenarios 3 and 4 with no recorded failures.

Nevertheless, false-positive detections occurred, particularly near walls and narrow corridors, and the system cannot distinguish people from other obstacles. Overall, the system demonstrates substantial elements of *Level 2 – Orientation and Interaction*, particularly independent localization using a known map, navigation and task execution. However, the absence of mapping prevents it from fulfilling all Level 2 descriptors, while dynamic obstacle detection provides only partial evidence of Level 3 – Context Awareness. Therefore, full achievement of Level 2 cannot be confirmed, and Level 3 has not been achieved.

6.4 Risk Analysis

The risk analyses in Tables 11 and 12 identify the main challenges associated with implementing the proposed AMR solution in each company.

As shown in Table 10, Company 1 would face challenges related to transitioning from manual transportation to autonomous operation, particularly regarding localization, obstacle detection, human–robot interaction, and fleet coordination.

Table 11. *Risk Analysis for AMR Implementation in Company*

Risk	Potential consequence	Possible mitigation
Localization errors	Incorrect navigation or positioning.	Improve localization accuracy and validate sensor performance.
Dynamic obstacle detection errors	Collisions or unnecessary stops.	Improve LiDAR detection and filtering.
Robot–human interaction	Potential collisions with warehouse workers.	Implement reliable human detection and safety-rated stopping mechanisms.
Multi-robot conflicts	Blocked routes or deadlocks.	Implement coordinated fleet management.
Operational transition	Disruption during the replacement of manual transport.	Introduce AMRs gradually and provide operator training.

In contrast, Table 12 shows that Company 2 would mainly face challenges related to adapting its existing automated transport system, including map limitations, obstacle detection, path replanning, and multi-robot coordination.

Table 12. *Risk Analysis for AMR Implementation in Company 2*

Risk	Potential consequence	Possible mitigation
Dependence on predefined maps	Navigation failures after environmental changes.	Implement map updates and eventually SLAM.
False obstacle detection	Unnecessary stops and reduced productivity.	Improve sensor accuracy and detection filtering.
Missed obstacle detection	Potential collisions with people or objects.	Introduce redundant sensing and validated safety mechanisms.
Path replanning failures	Robots remain blocked or fail to complete tasks.	Improve replanning and recovery strategies.
Multi-robot coordination	Traffic conflicts and operational delays.	Implement centralized fleet coordination.

Both implementations require reliable perception and coordination mechanisms, although their specific challenges differ according to their current level of automation. The proposed mitigation measures represent recommendations for future deployment and have not all been validated in the simulation.

6.5 Human–Robot Shared Environment

The proposed system addresses human–robot coexistence, where people and robots share the same workspace without performing shared tasks (Malik et al., 2019). Robots can respond to potential obstacles by reducing speed, stopping or replanning their routes. However, reliable human identification and prediction of human behavior are not implemented.

Although these mechanisms provide a basic safety response, they do not guarantee safe human–robot interaction or establish compliance with ISO 3691-4:2023.

6.6 Deployment Considerations

Real-world deployment would require physical testing, improved obstacle perception, validated safety mechanisms and integration with existing warehouse operations. A site-specific risk assessment and validation under different operating conditions would also be necessary before industrial use.

7 Conclusions

This project developed and evaluated a multi-robot AMR simulation for autonomous material handling in a warehouse environment. The system integrated Particle Filter localization, LiDAR-based environment perception, A* path planning, task allocation, and dynamic obstacle detection. Two robots were able to execute multiple pickup-and-delivery tasks while coordinating their routes and responding to detected obstacles.

The simulation results demonstrated that Particle Filter localization generally improved positioning accuracy compared with odometry alone. The robots also completed their assigned tasks in the tested scenarios, including dynamic obstacle experiments, where all waypoints were reached without recorded robot failures. However, performance was affected by localization errors under high-noise conditions and false-positive obstacle detections, particularly near walls and narrow corridors.

Compared with the original objectives, most of the proposed capabilities were implemented and evaluated. Nevertheless, human identification was not achieved because the system cannot distinguish people from other obstacles. Furthermore, the use of a predefined map means that SLAM and autonomous map generation were outside the project scope. These limitations prevent the system from demonstrating the full range of capabilities associated with the targeted autonomy level.

According to the Franke et al. (2026) framework, the proposed system demonstrates several capabilities associated with **Level 2 – Orientation and Interaction**, particularly localization, navigation, and task execution. However, since mapping is not implemented, full achievement of this level cannot be confirmed. The system also shows partial context-awareness capabilities, but it does not meet the requirements for Level 3 – Context Awareness.

Overall, the project demonstrates the feasibility of coordinating multiple AMRs for warehouse material handling within a simulated environment. Further development should focus on improving perception reliability, implementing human identification, and supporting adaptation to changes in the environment. Physical testing would also be necessary to evaluate the system under real operating conditions.

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